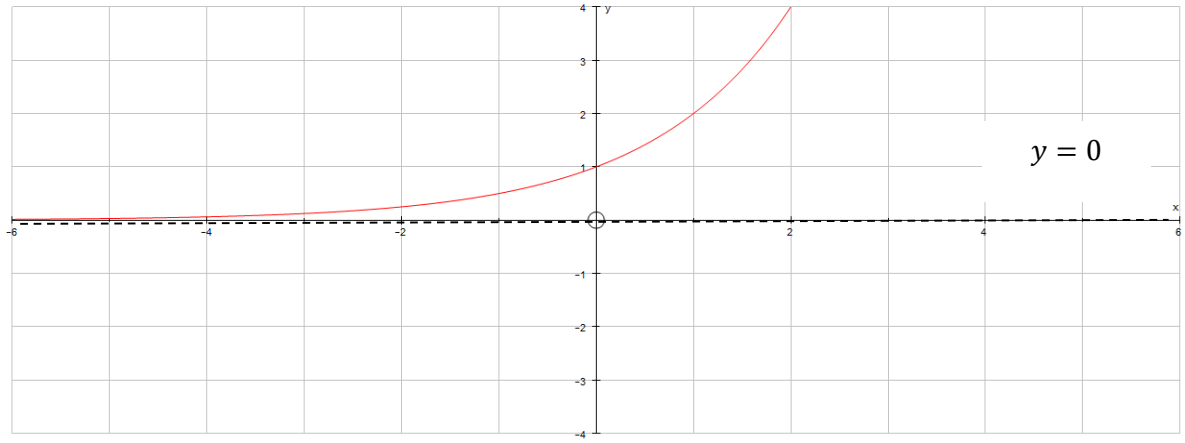
E-Classroom	**Name: Memorandum**					
	Gr 10		**Date:**		**Time:**	$2\frac{1}{2}$ **HR**
CAPS Reference	**Functions**					
Topic	**Exponential Function**					

2.1 $f(x) = 2^x$

Asymptote

$y = 0$

x	-2	-1	0	1	2
2^x	$\frac{1}{4}$	$\frac{1}{2}$	1	2	4



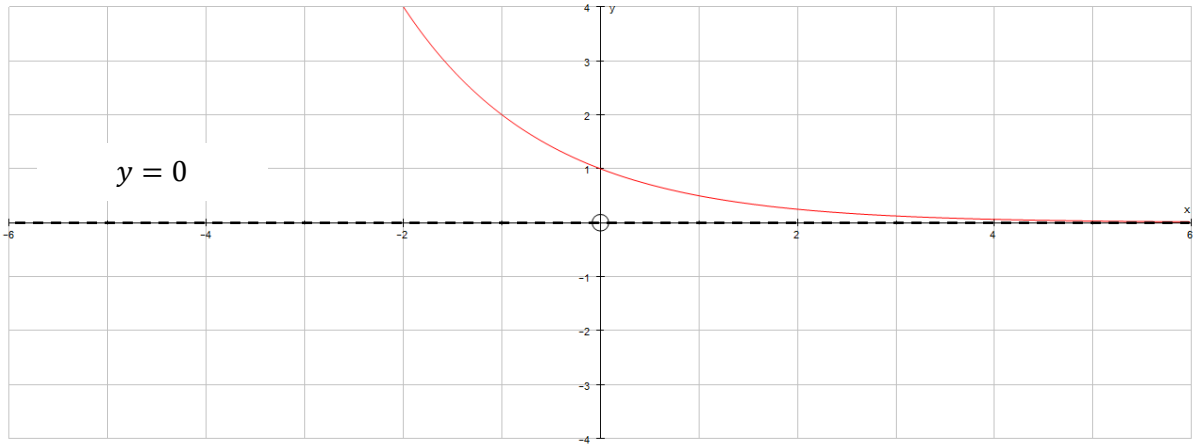
a) Range: $y \in (0; \infty)$
b) $y = 0$
c) $x \in R$
d) $h(x) = f(x) - 1$
$= 2^x - 1$

2.2 $g(x) = \left(\frac{1}{2}\right)^x$

Asymptote

$y = 0$

x	-2	-1	0	1	2
$\left(\frac{1}{2}\right)^x$	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$



a) Range: $y \in (0; \infty)$
b) $y = 0$
c) $x \in R$
d) $h(x) = g(x) + 1$

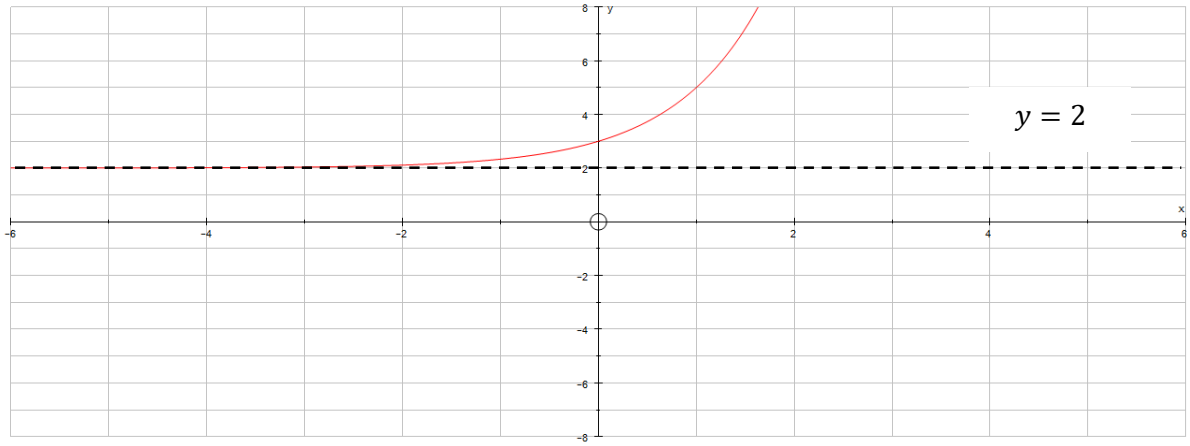
$= \left(\frac{1}{2}\right)^x + 1$

2.3 $f(x) = 3^x + 2$

Asymptote

$y = 2$

x	-2	-1	0	1	2
$3^x + 2$	$\frac{19}{9}$	$\frac{7}{3}$	3	5	11



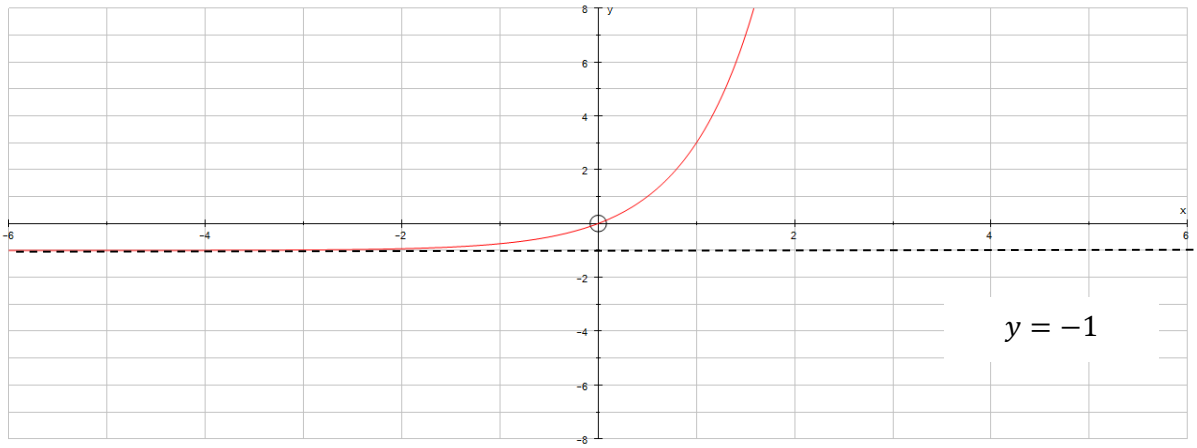
a) Range: $y \in (2; \infty)$
b) $y = 2$
c) $x \in R$
d) $h(x) = -f(x)$
$= -(3^x + 2)$
$= -3^x - 2$

2.4 $g(x) = 4^x - 1$

Asymptote

$y = -1$

x	-2	-1	0	1	2
$4^x - 1$	$-\frac{15}{16}$	$-\frac{3}{4}$	0	3	15



a) Range: $y \in (-1; \infty)$
b) $y = -1$
c) $x > 0;\ x \in R$
d) $h(x) = g(x) - 2$
$= 4^x - 1 - 2$
$= 4^x - 3$

4.1 $f(x) = a.2^x + q$

Step 1: Substitute the asymptotes $y = 3$

$y = a.2^x + 3$

Step 2: Substitute the given coordinate $(0; 2)$

$y = a.2^x + 3$

$2 = a.2^0 + 3$

$2 - 3 = a$

$-1 = a$

Step 3: Substitute a in to the equation.

$y = -2^x + 3$

$$\boldsymbol{f(x) = -2^x + 3}$$

4.2 $f(x) = a.2^x + q$

Step 1: Substitute the asymptotes $y = 2$

$y = a.2^x + 2$

Step 2: Substitute the given coordinate $(0; 3)$

$y = a.2^x + 2$

$3 = a.4^0 + 2$

$3 - 2 = a$

$1 = a$

Step 3: Substitute a into the equation.

$y = 2^x + 2$

$$\boldsymbol{f(x) = 2^x + 2}$$

4.3 $f(x) = a.3^x + q$

Step 1: Substitute the asymptotes $y = -1$

$y = a.3^x - 1$

Step 2: Substitute the given coordinate $(0; -2)$

$y = a.3^x - 1$

$-2 = a.3^0 - 1$

$-2 + 1 = a$

$-1 = a$

Step 3: Substitute a in to the equation.

$y = -3^x - 1$

$$\boldsymbol{f(x) = -3^x - 1}$$

4.4 $f(x) = a.\left(\frac{1}{2}\right)^x + q$

Step 1: Substitute the asymptotes $y = 2$

$y = a.\left(\frac{1}{2}\right)^x + 2$

Step 2: Substitute the given coordinate $(0; 3)$

$y = a.\left(\frac{1}{2}\right)^x + 2$

$3 = a.\left(\frac{1}{2}\right)^0 + 2$

$3 - 2 = a$

$1 = a$

Step 3: Substitute a into the equation.

$y = \left(\frac{1}{2}\right)^x + 2$

$$\boldsymbol{f(x) = \left(\frac{1}{2}\right)^x + 2}$$